

Efficient Visualization of Large-Scale Unstructured Grid Data through Importance-Based Hybrid Reduction

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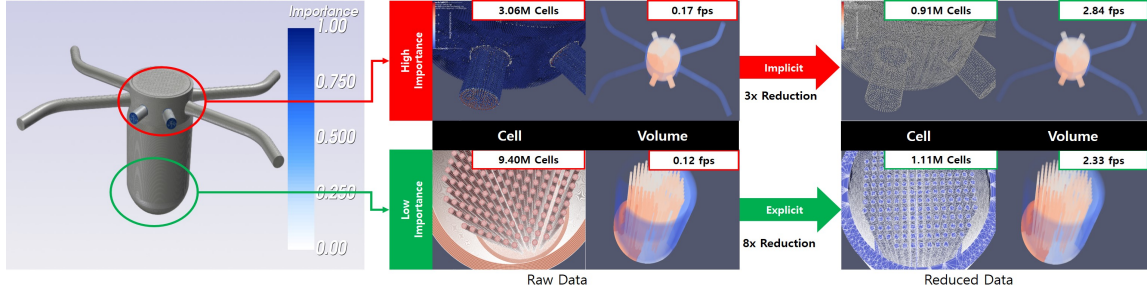


Figure 1: Hybrid reduction: importance map → original → hybrid result (3–8× size ↓, ≈ 99% field accuracy).

Abstract

For large CFD/FEA unstructured grids, we propose an importance-driven hybrid reduction: explicit decimation for low-importance regions and learning-based reduction for salient subvolumes, achieving 3–8× size reduction with ≈ 99% field accuracy and higher Vol. FPS at matched cells.

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1 Introduction

Modern CFD/FEA datasets contain tens to hundreds of millions of cells, overwhelming visualization pipelines and turning view changes into minute-scale waits. Existing approaches fail to balance quality and performance: uniform decimation loses critical features, while learning methods struggle with geometric accuracy. Recent advances in error-bounded compression [Di et al. 2025] mitigate trade-offs. Mesh simplification preserves topology but ignores field importance [Kim et al. 2024], while neural compression achieves high ratios but lacks interpretability [Weiss et al. 2022].

Our hybrid reduction framework combines both approaches' strengths through importance-driven selection. Figure 1 demonstrates this. Full-domain implicit reduction is impractical for our

targets due to VRAM and training-time budgets, so we preserve critical regions while aggressively coarsening unimportant areas.

2 Method

Our pipeline (Figure 2) consists of two components: (1) ensemble importance analysis that identifies critical regions by combining metrics, and (2) hybrid reduction that applies appropriate simplification to each region based on importance.

2.1 Ensemble Importance Analysis

Single importance metrics can miss specific features, so we combine signals from different perspectives to identify critical regions, as shown in the left portion of Figure 2. We fuse complementary signals: PCA (global variance), Random Forest and SHAP (model-based importance for nonlinear relations), mutual information (statistical dependence), and LASSO (L1-sparse feature selection).

We normalize each signal to [0,1] and adopt equal-weight averaging ($w_m = 1$) for stability and transparency; ablations with rank-based fusion and L1-regularized meta-weighting yielded near-identical region rankings, so we retain $w_m = 1$ while using them to validate robustness. Spatial variations are captured through gradient $\nabla(f * G_\sigma)$ and Laplacian $\Delta(f * G_\sigma)$ operators, highlighting rapid changes and extrema.

2.2 Hybrid Reduction Pipeline

2.2.1 Explicit Reduction for Low-Importance Regions ($I < \tau$). Low-importance regions ($I < \tau$) undergo geometry-based explicit simplification to reduce computational and memory overhead, as illustrated in the lower path of Figure 2. The retention ratio follows a linear schedule based on normalized importance (very low importance retains 15%, near threshold retains 50%). The pipeline proceeds through adaptive downsampling, surface extraction, and quality-constrained simplification/remeshing. Physical fields are

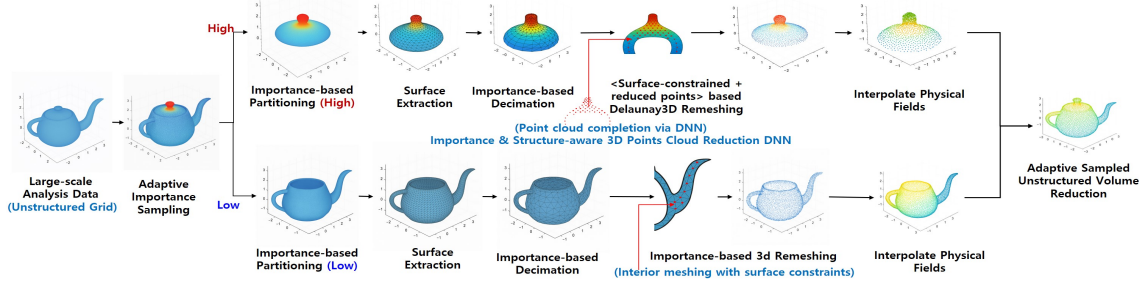


Figure 2: Conceptual hybrid pipeline: importance analysis → implicit volume for salient regions → explicit decimation for the rest → unified reconstruction for interactive rendering.

consistently transferred to the reduced mesh, with barycentric interpolation within tetrahedra ensuring field continuity.

2.2.2 Implicit Reduction for High-Importance Regions ($I \geq \tau$). For high-importance regions ($I \geq \tau$), we employ a learning-based implicit representation to preserve boundaries and fine structures while the remainder is simplified explicitly. The architecture uses a 1D convolutional encoder–decoder (64–512 channels) with hybrid pooling (max+average) and LayerNorm/Dropout regularization. We adopt per-asset optimization (no train/test split, fixed hyperparameters). Through joint geometry–field learning, the model optimizes a composite loss combining: (i) geometric accuracy (Chamfer distance, directional consistency), (ii) field fidelity (L2 norm), and (iii) conservation constraints for boundary and physical properties. The critical points derived from this process are used alongside surface information for final volume remeshing. Adaptive sampling increases density in high-importance areas to enhance gradient (∇) preservation near boundaries. The hybrid grid is generated through surface-constrained tetrahedralization. We reuse cell-based barycentric interpolation from Section 2.2.1 to enhance accuracy.

3 Results and Evaluation

Table 1: Evaluation Results with Volumetric Metrics

Dataset	Cells	Reduction Ratio	Vol. FPS	CNRMSE (%)	nVol. SDF RMSE
<i>SMR High-Importance</i>					
Original	3.06M	1×	0.17	-	-
Uniform (Baseline)	0.91M	3.36×	0.75	0.80	0.001294
Hybrid-Implicit (Ours)	0.91M	3.36×	2.84	0.11	0.000263
<i>SMR Low-Importance</i>					
Original	9.4M	1×	0.12	-	-
Uniform (Baseline)	1.11M	8.46×	0.47	0.56	0.002003
Hybrid-Explicit (Ours)	1.11M	8.46×	2.33	0.44	0.000391

SMR: Small Modular Reactor simulation dataset. All comparisons match cells/VRAM.

Vol. SDF-RMSE measures signed-distance error over interior and surface.

CNRMSE = $\text{RMSE}_f / (\max f - \min f)$ (f : field values).

nVol. SDF-RMSE = $(1/L) \cdot \sqrt{\sum w_i (\phi(x_i) - \hat{\phi}(x_i))^2 / \sum w_i}$, $w_i = e^{-\alpha |\phi(x_i)|}$;

ϕ : ground truth SDF, $\hat{\phi}$: reduced SDF, L : bbox diagonal, $\alpha = \ln(2)/(0.05L)$, $N = 100k$.

We evaluate our hybrid approach using CNRMSE (normalized RMSE of field values) and a normalized volumetric SDF-RMSE (3D-sampled signed distance differences with exponential depth weighting and diagonal normalization). We also report Vol. FPS across multiple scalar fields. These volumetric metrics provide a more comprehensive assessment than surface metrics.

On the SMR High-Importance dataset, at 0.91M cells our hybrid-implicit method achieves 2.84 Vol. FPS versus 0.75 for uniform (baseline) ($\approx 3.8\times$), while reducing CNRMSE from 0.80% to 0.11% and nVol. SDF-RMSE from 1.294×10^{-3} to 2.63×10^{-4} . At matched cell counts, Vol. FPS increases because rendering cost scales with ray-cell intersections and memory locality. On the SMR Low-Importance dataset, at 1.11M cells our hybrid-explicit method achieves 2.33 Vol. FPS versus 0.47 for uniform (baseline) ($\approx 5.0\times$), while reducing CNRMSE from 0.56% to 0.44% and nVol. SDF-RMSE from 2.003×10^{-3} to 3.91×10^{-4} . The SMR High-Importance dataset exhibits higher spatial variation, requiring conservative reduction to preserve accuracy. Uniform decimation fails to adapt—yielding higher CNRMSE for high-variation regions (0.80% vs 0.56%)—our hybrid applies importance-aware reduction: minimal simplification for high-importance areas and aggressive coarsening for low-importance regions, achieving superior accuracy across both datasets. Full-domain implicit/explicit baselines were infeasible at our scale because they exceeded practical VRAM and optimization-time budgets for large unstructured grids; CNRMSE gains vs Uniform on implicit (high-I) and explicit (low-I) subsets proxy global behavior and support the hybrid; full-domain baselines are future work.

4 Conclusion

This research combines ensemble importance analysis with hybrid reduction to enable interactive volume rendering on large unstructured grids within practical memory budgets. We demonstrate preservation of critical structures and improved visualization performance, maintaining a balance between quality and efficiency.

Acknowledgments

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